

Chapter 11

Object Detection

IMP302m

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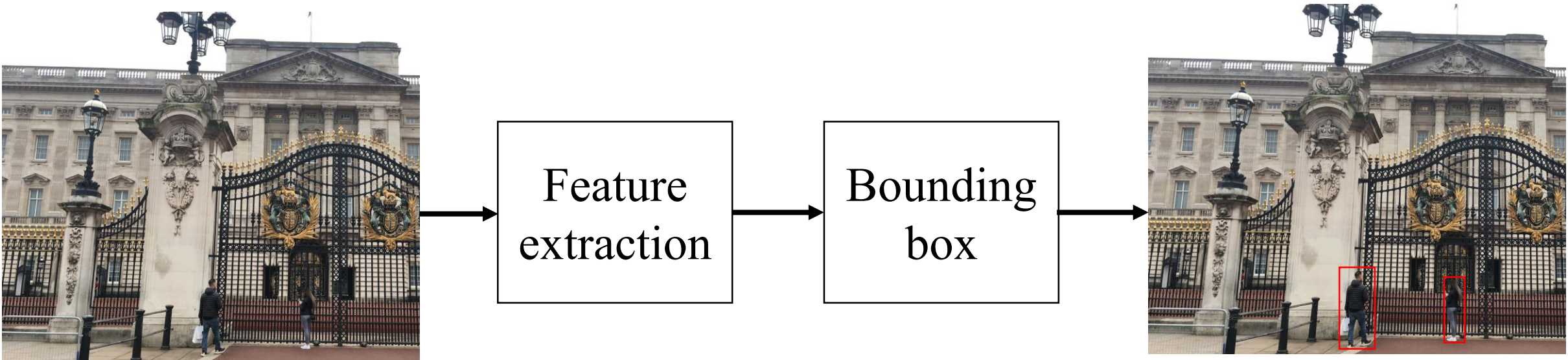
- 11.1 Traditional methods
- 11.2 Machine learning based methods

Fundamentals

- Object detection is a computer vision technique that identifies and locates objects within an image.
- It combines image classification and object localization.
 - Discussion: What type of classification is used in object detection?
- Common applications: widely used in digital era
 - autonomous vehicles, facial recognition, and surveillance

Fundamentals

- A simple diagram showing the process



Fundamentals – Evaluation metrics

- INTERSECTION OVER UNION (IoU): designed to measure the detection quality by calculating the difference between the ground truth annotations and the predicted bounding boxes

$$IoU = \frac{\text{Area of union}}{\text{Area of intersection}}$$

- What does it mean if IoU value is closed to 1?

Fundamentals – Evaluation metrics

- MEAN AVERAGE PRECISION (mAP): the mean average precision of all K classes
 - The average precision (AP) is derived from the precision-recall curve, calculated for all unique recall levels
- MEAN AVERAGE RECALL (mAR): the mean value of the RAs for all K classes
 - the average recall (AR) also represents a numerical metric to compare the detector's efficiency

Fundamentals – Datasets

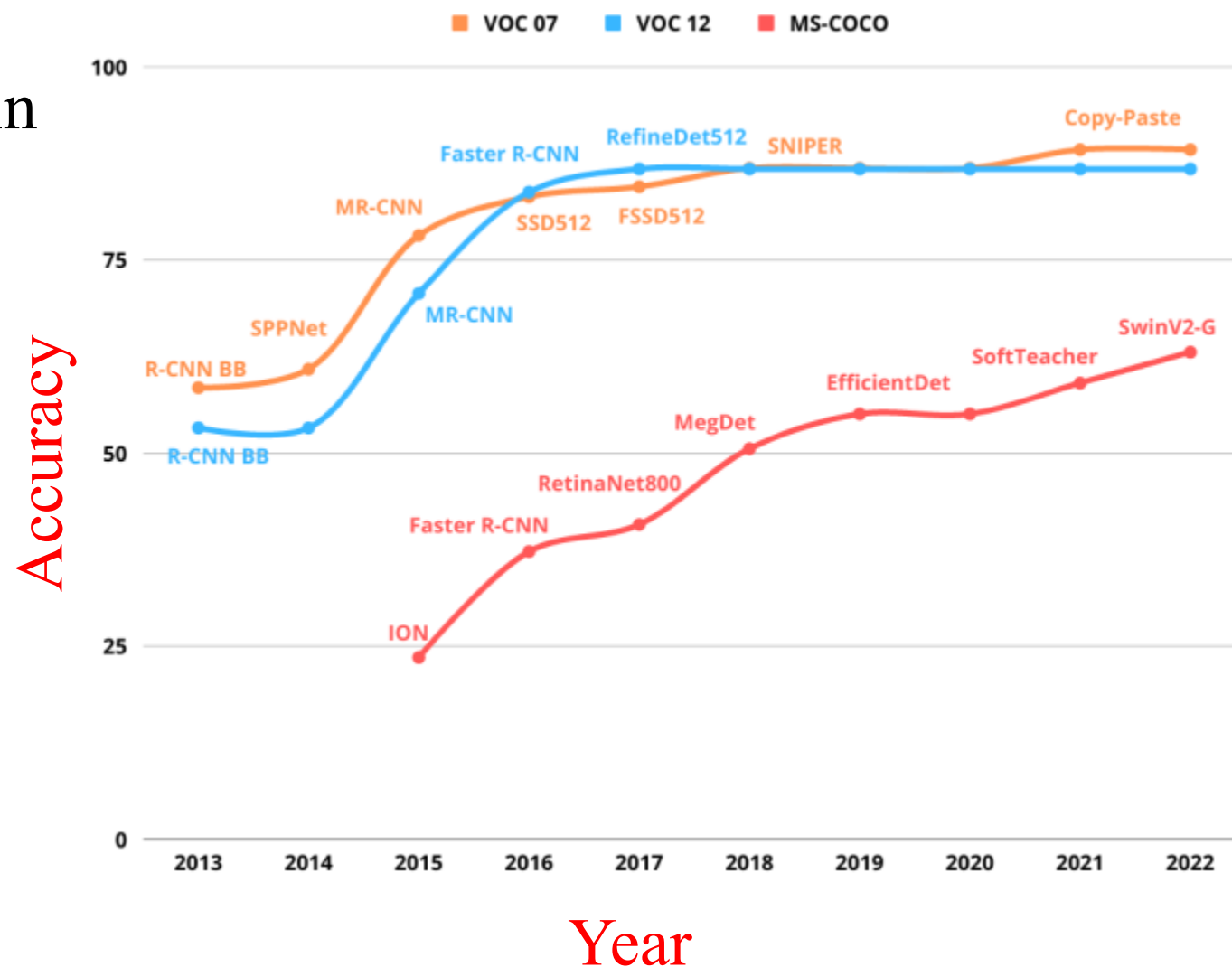
- PASCAL Visual Object Classification (PASCAL VOC)
 - About 10,000 training and validation images with objects and bounding boxes
 - 20 different categories
- The common Objects in COntext (COCO)
 - About 200,000 training, validation and testing images
 - 80 categories

Fundamentals – Datasets

- The ImageNet Large Scale Visual Recognition Challenge (ILSVRC)
 - About 14 million of training, validation and testing images
 - 200 categories
- Open Images
 - About 9 millions of training, validation and testing images
 - 600 categories

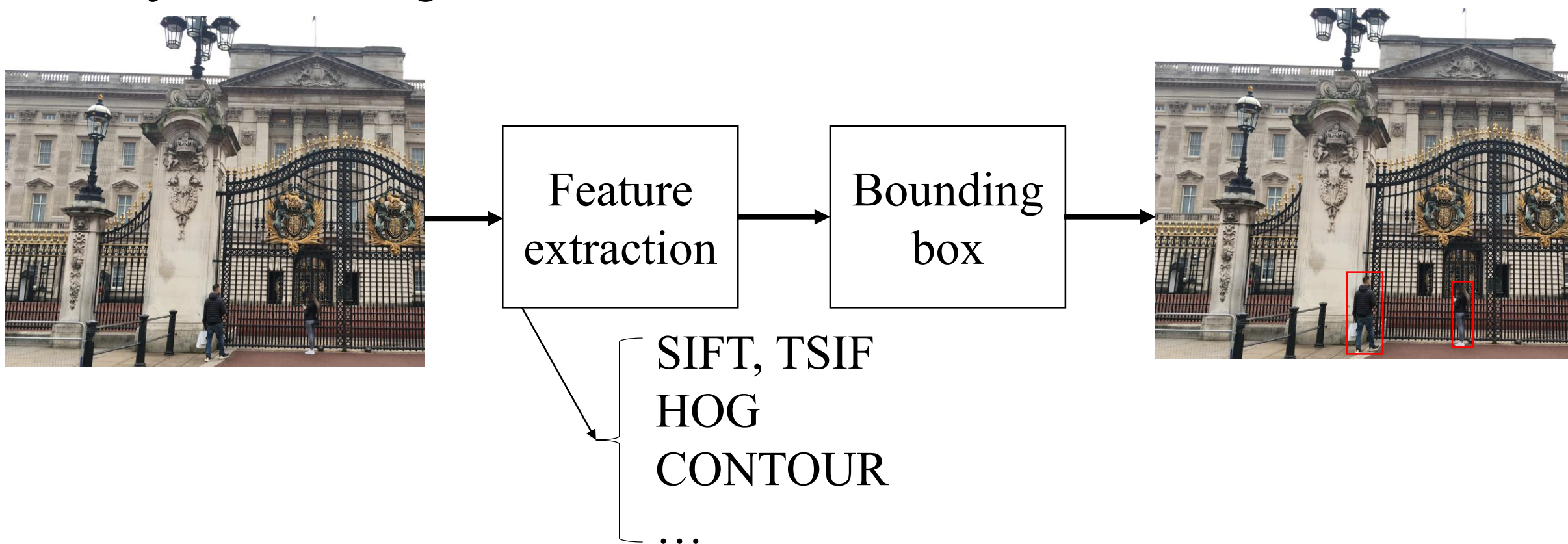
Fundamentals – Datasets

- Accuracy evolution in the main object detection benchmarks



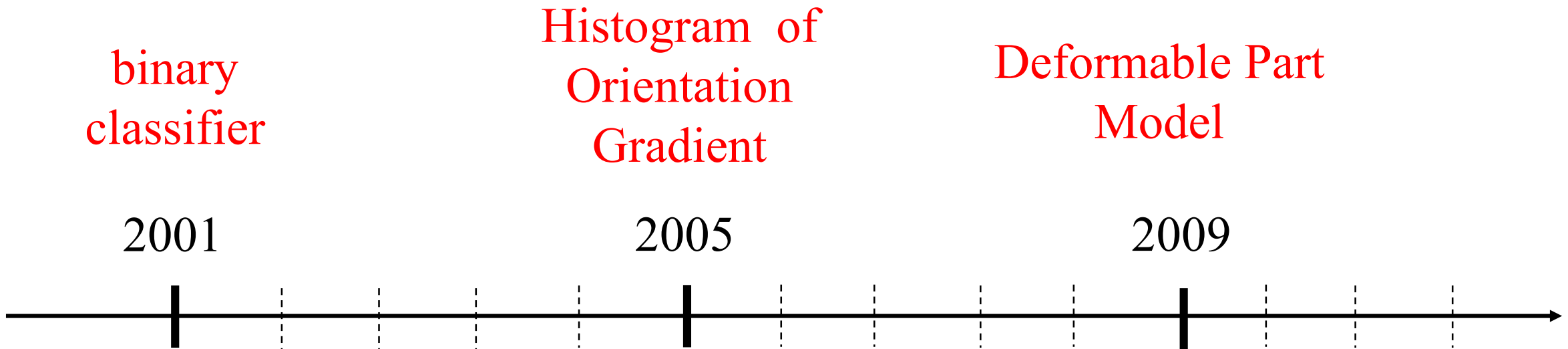
11.1 Traditional methods

- Traditional methods utilized traditional feature extraction to detect object in an image



11.1 Traditional methods

- Brief development timeline of traditional methods

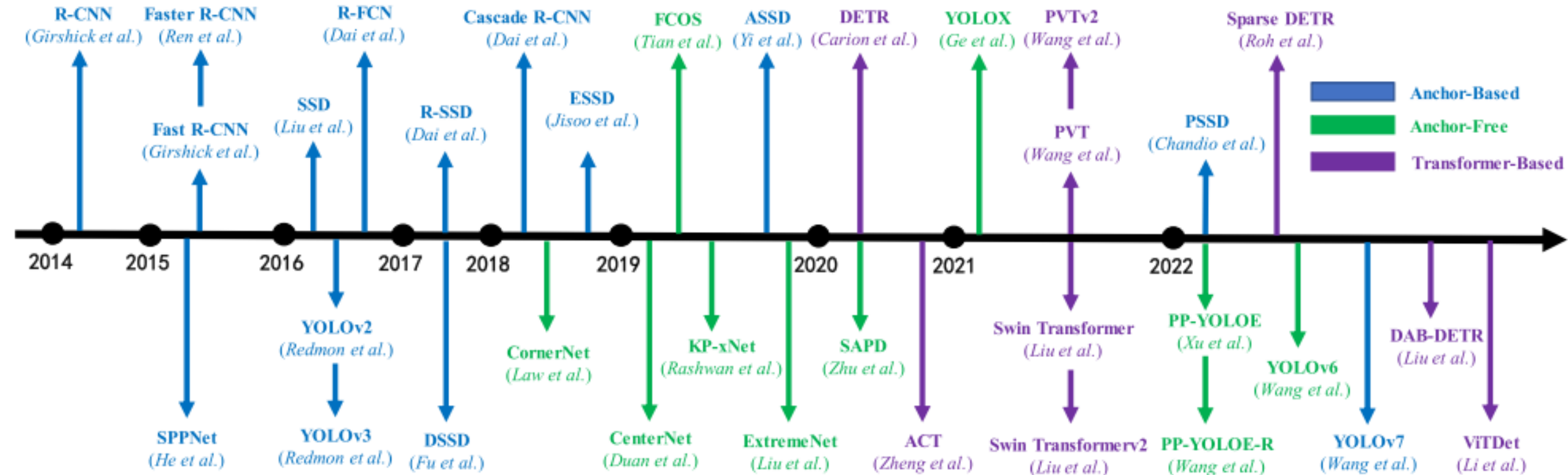


11.1 Traditional methods

- It can be divided into three steps
 - the selection of informative places
 - features extraction
 - object categorization
- Objects might appear in various visual places and have different sizes and aspect ratios → a linear kernel with many scales to scan the entire image → costs high computation and generates many insignificant candidates

11.2 Machine learning based methods

- Brief development timeline of machine learning based methods

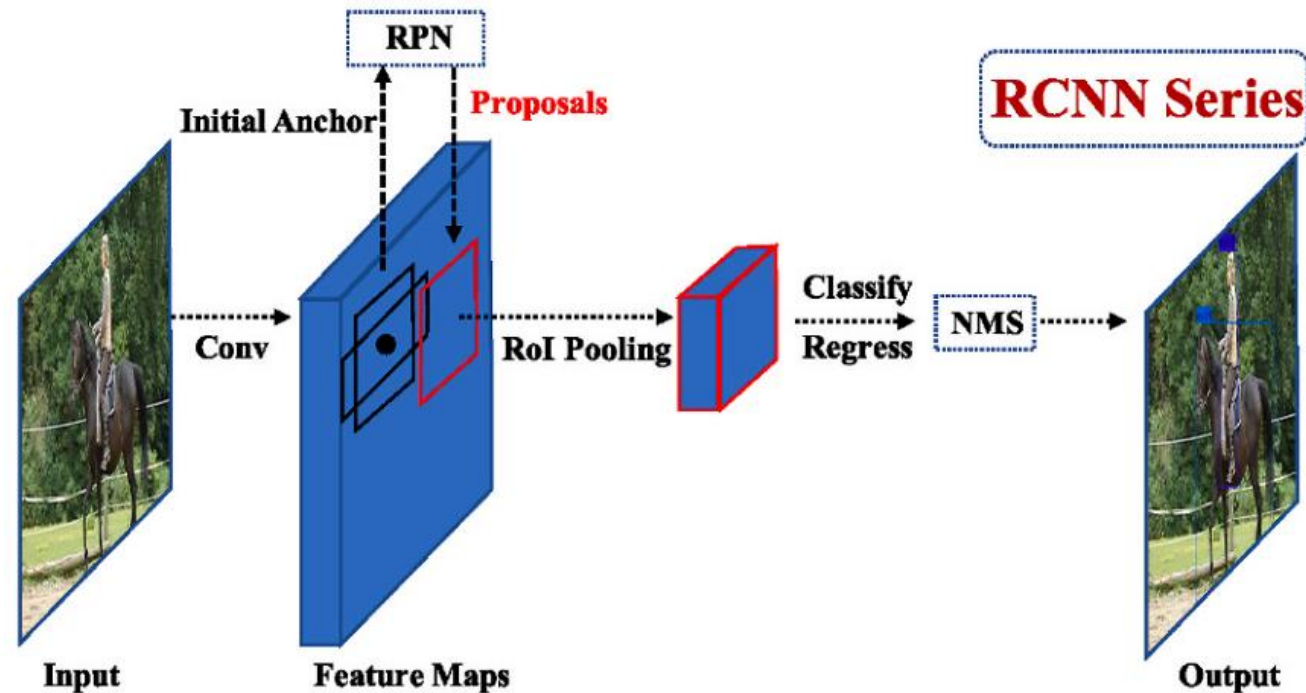


11.2 Machine learning based methods - Anchor-based methods

- Anchor-based methods are the methods which predefine a set of dense anchor boxes of different scales and to differentiate these anchor boxes into positive and negative samples based on a certain labeling assignment strategy → position offset for positive samples are predicted

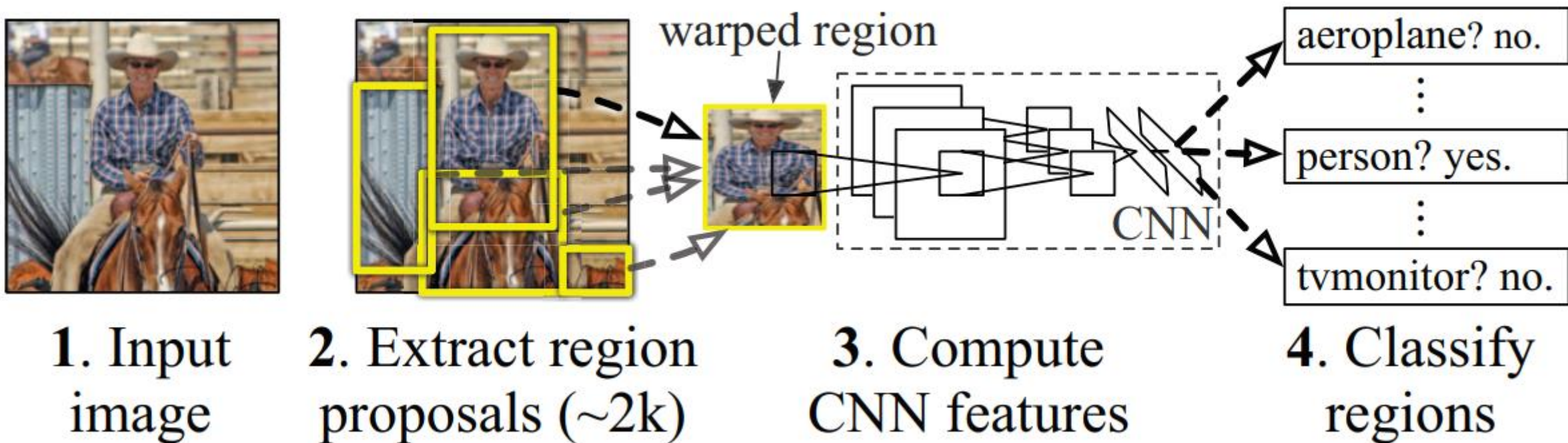
11.2 Machine learning based methods - Anchor-based methods

- **RCNN series**
- Generates a large number of candidate boxes for the input image through a selective search algorithm
- Inputs them into the convolution backbone to extract features
- Uses SVM and linear regressor for classification and location fine-tuning



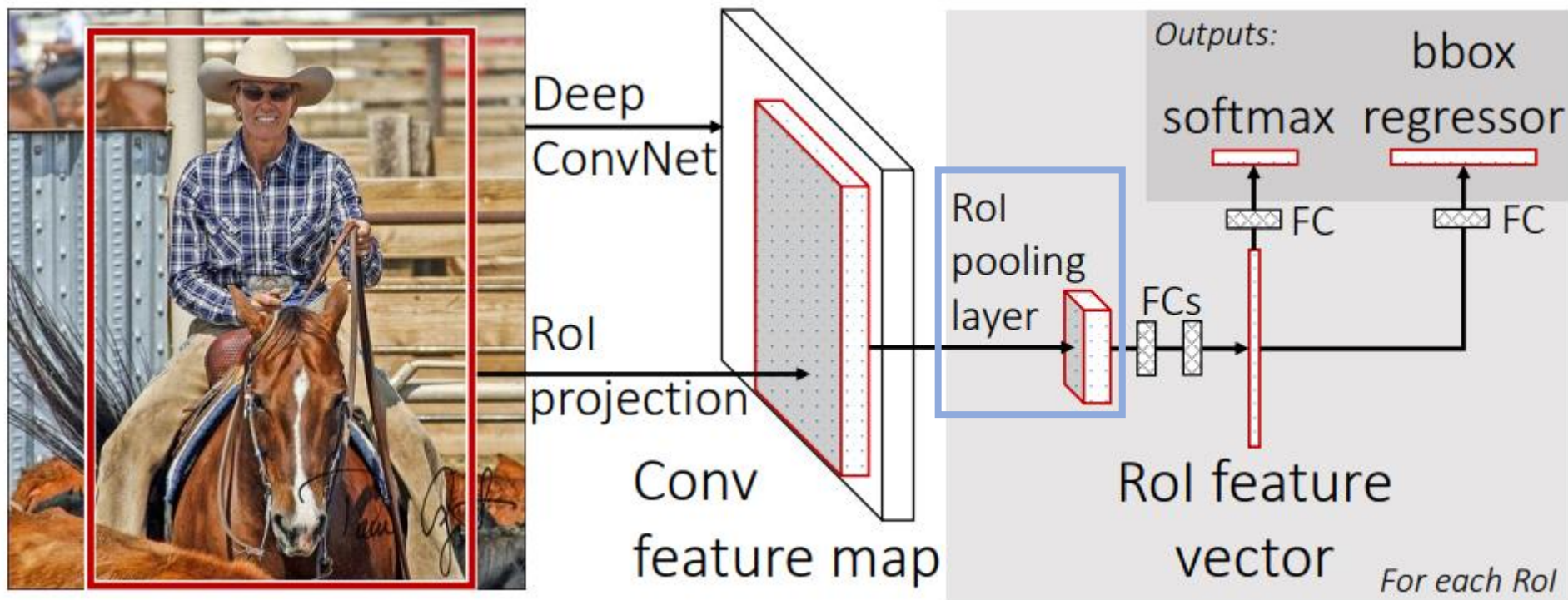
11.2 Machine learning based methods - Anchor-based methods

- **RCNN series**
- R-CNN (Regions with CNN features)



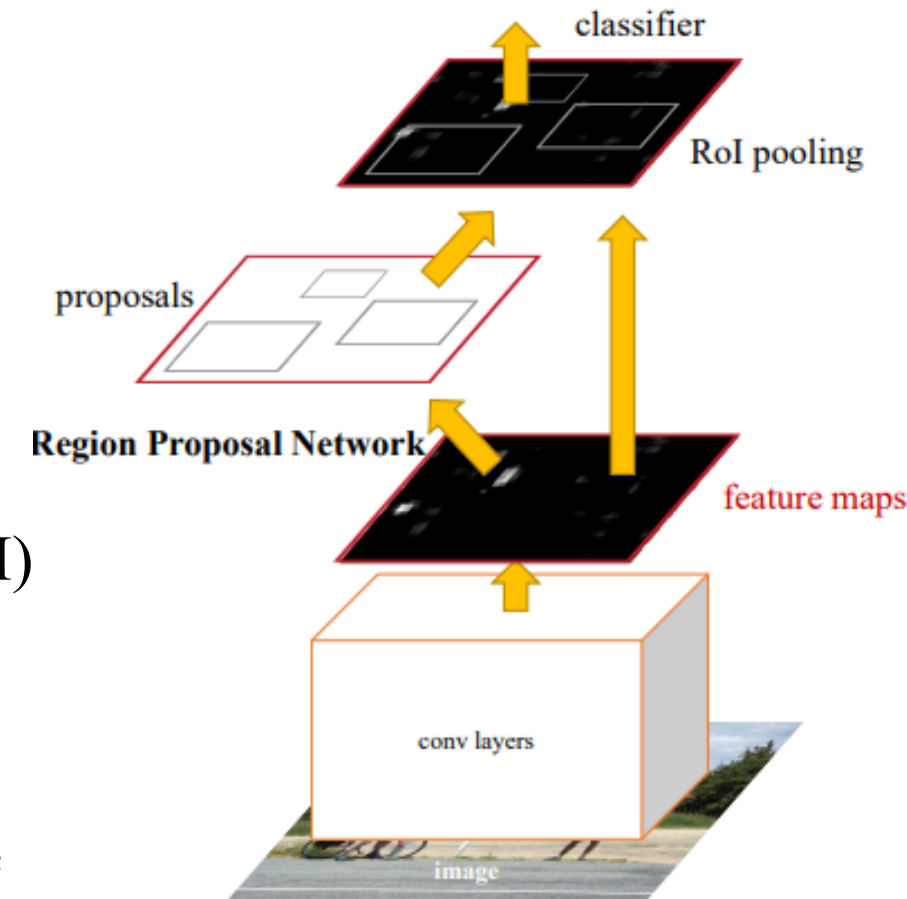
11.2 Machine learning based methods - Anchor-based methods

- **RCNN series**
- Fast RCNN: improves the accuracy by 11.5%, runs more than 200 times faster in comparison with R-CNN



11.2 Machine learning based methods - Anchor-based methods

- **RCNN series**
- **Faster RCNN**
 - generate anchor candidates through a Region Candidate network (RPN) that shares features with the detection network
 - RoI pooling layer maps the foreground anchor (RoI) to a fixed-size feature map
- Puts candidate generation, feature extraction, and final classification and regression into the same convolutional neural network → increase speed



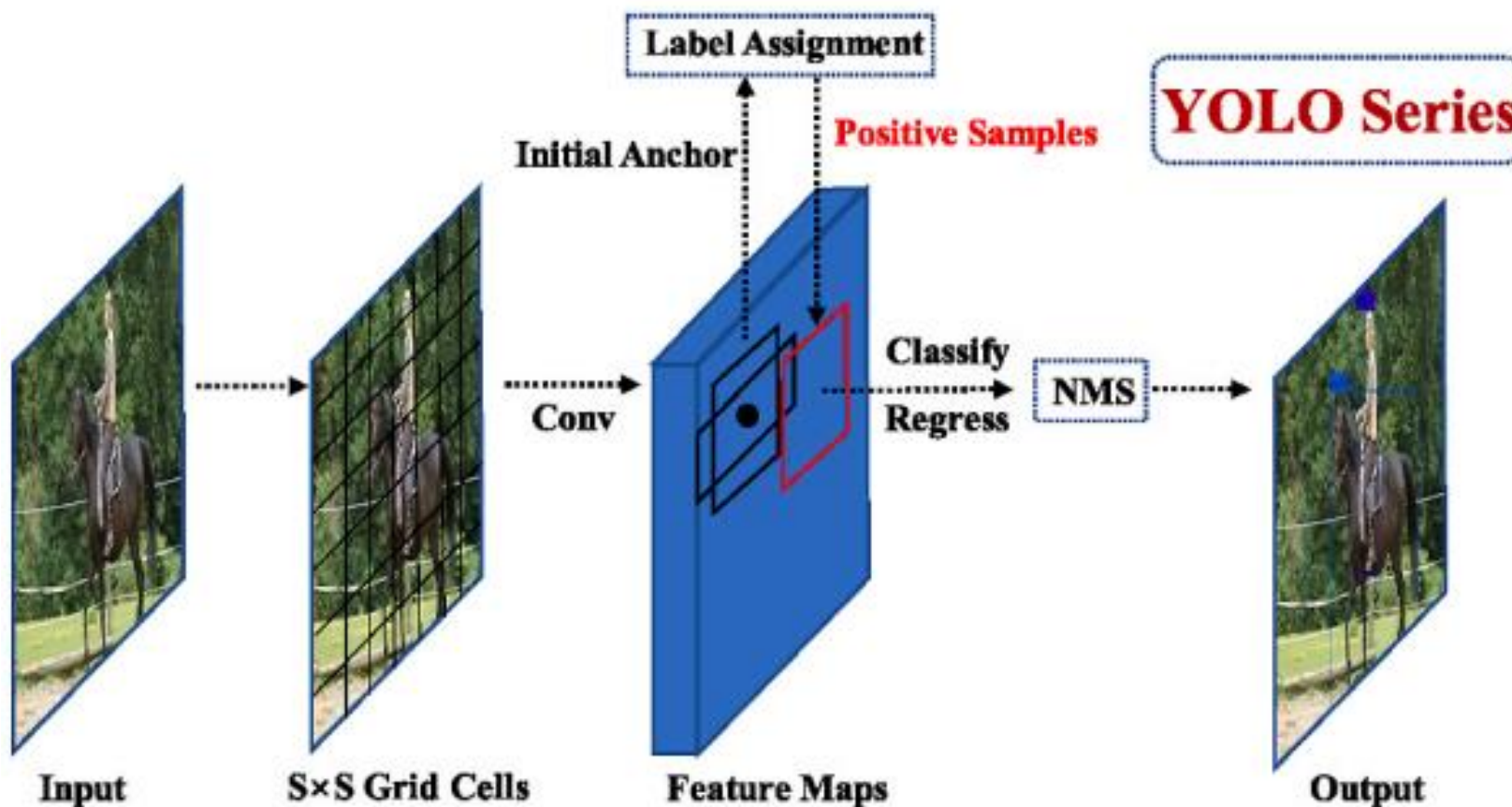
11.2 Machine learning based methods - Anchor-based methods

• RCNN series

R-CNN	- Simple to use - Application of CNN for classification
Fast R-CNN	- Faster than R-CNN - Introduce RoIU pooling layer for mapping various input sizes - Multi-task model by integrating bounding box regression along with classification - The ability to use an Region Proposal Network and CNN pattern
Faster R-CNN	- With RPN instead of selective search, generating regional proposals requires significantly less time - Introducing anchor boxes - Multi-task loss - High performance in terms of accuracy - End-to-end learning

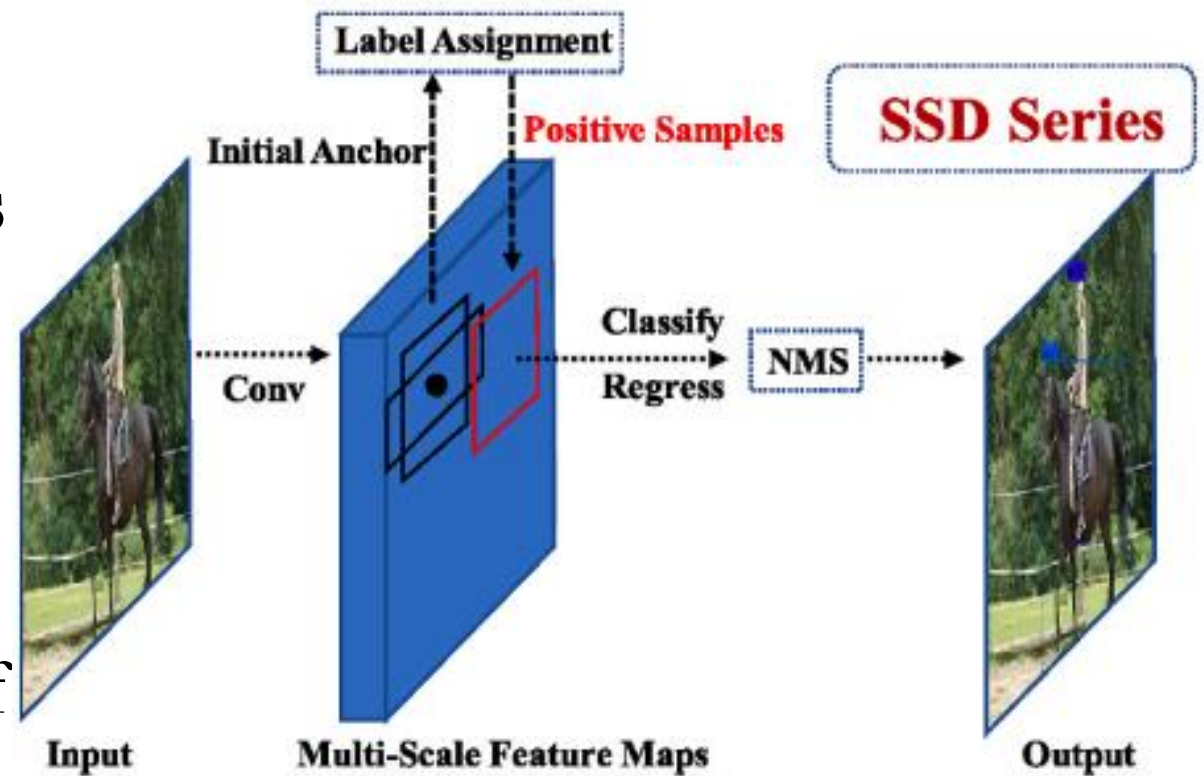
11.2 Machine learning based methods - Anchor-based methods

- YOLO (you only look once) series



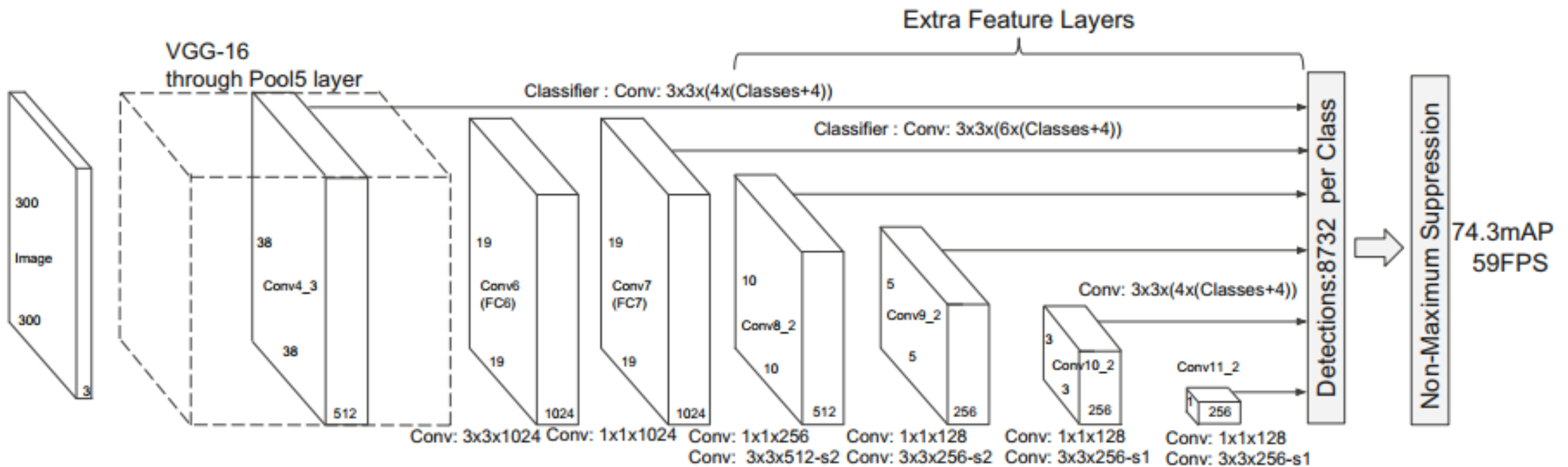
11.2 Machine learning based methods - Anchor-based methods

- **SSD (Single-Shot Detector) series**
- The core of SSD is predicting category scores and box offsets for a fixed set of default bounding boxes using small convolutional filters applied to feature maps.
- High detection accuracy because of predictions of different scales from feature maps of different scales, and explicitly separate predictions by aspect ratio



11.2 Machine learning based methods - Anchor-based methods

- SSD model

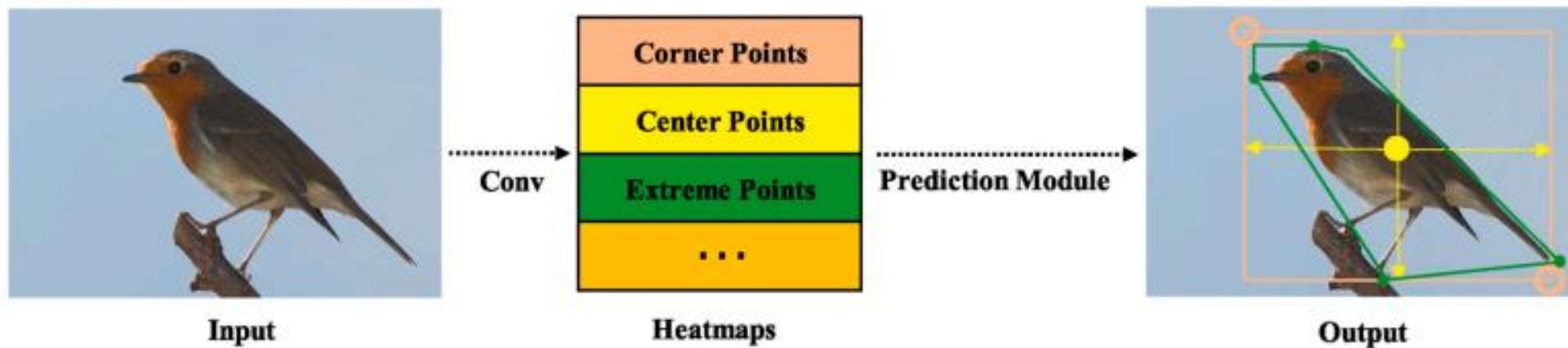


11.2 Machine learning based methods - Anchor-free methods

- Limitations of anchor-based methods
 - The design of hyperparameters such as the size and aspect ratio of the anchor box has a great impact on the performance of the detection model, and the anchor box designed solely by artificial prior is easy to lead to poor generalization ability of the model.
 - Most of the generated dense anchor boxes are negative samples, which will cause the problem of class imbalance between positive and negative samples.
 - The square anchor box of positive samples still has the problems of positioning ambiguity and background feature interference.

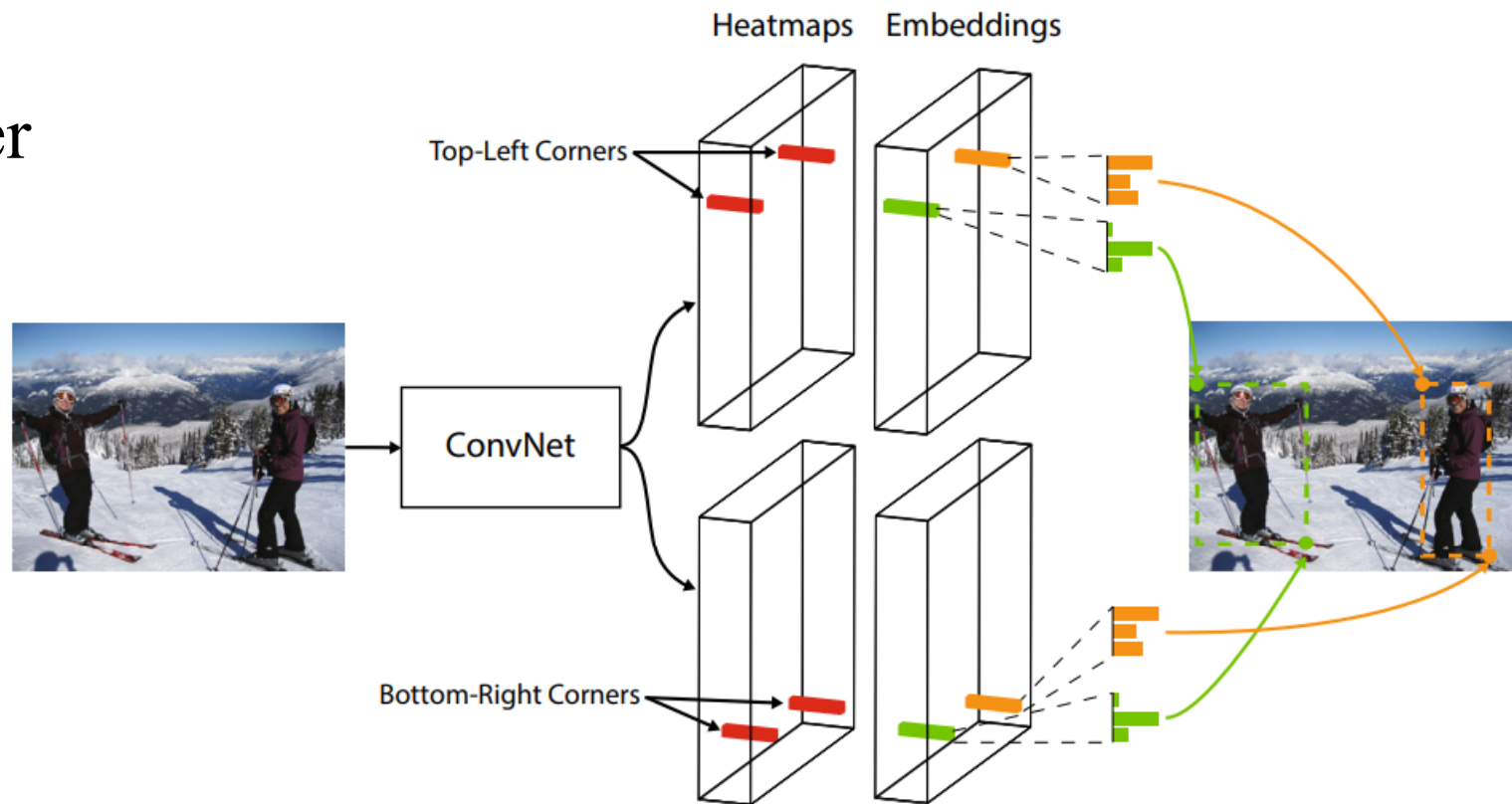
11.2 Machine learning based methods - Anchor-free methods

- Key-point-based methods: achieve anchor-free object detection by representing prediction boxes with sparse key corners



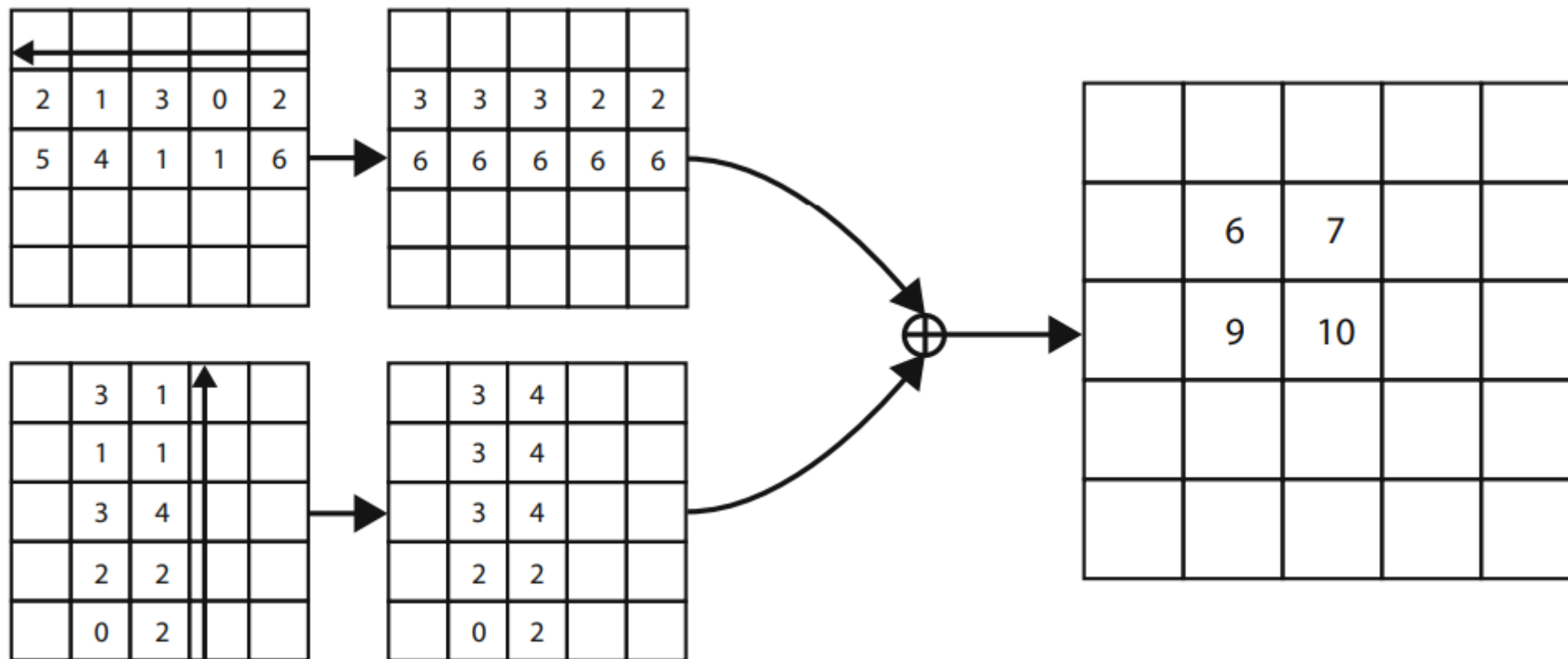
11.2 Machine learning based methods - Anchor-free methods

- **CornerNet**
- Novel component is corner pooling, a new type of pooling layer that helps a convolutional network better localize corners of bounding boxes
- Discussion: corners can help to detection in compared with bounding box centers or proposals?



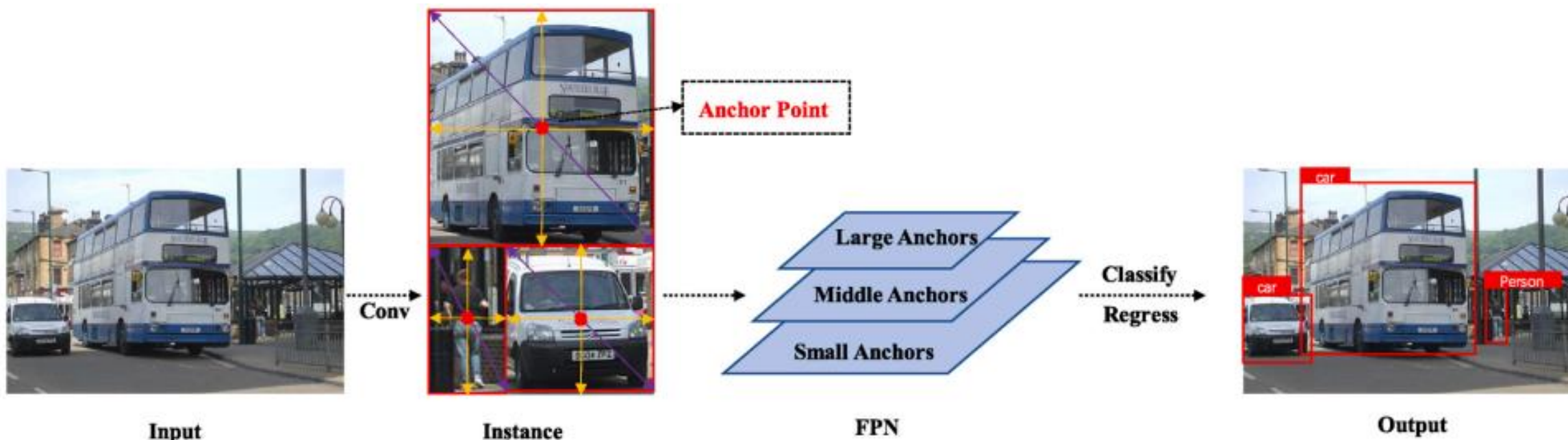
11.2 Machine learning based methods - Anchor-free methods

- **CornerNet**
- The top-left corner pooling layer



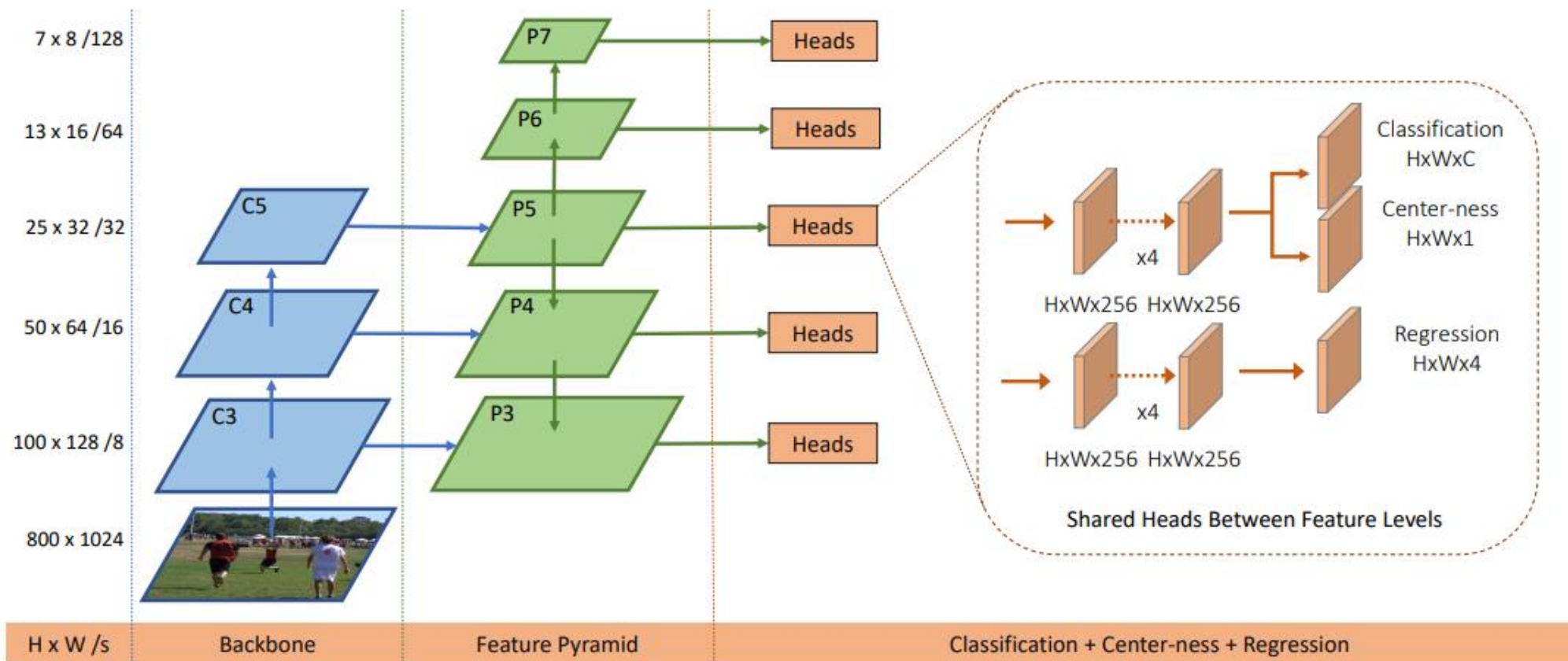
11.2 Machine learning based methods - Anchor-free methods

- Anchor-point-based methods



11.2 Machine learning based methods - Anchor-free methods

- FCOS (Fully Convolutional One-Stage Object Detector)

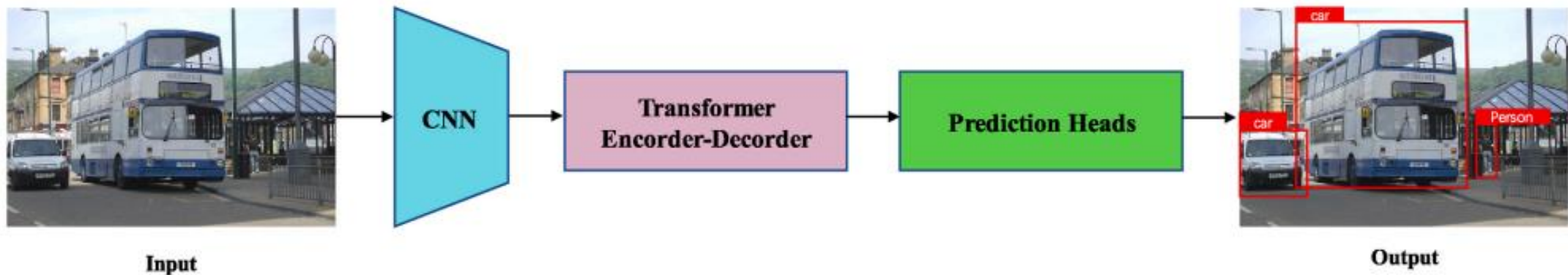


11.2 Machine learning based methods - Transformer-based methods

- Convolutional neural networks extract features by sliding independent convolutional windows → the model unable to effectively obtain global feature information
- Transformer can capture long-range dependencies on the object and easily obtain global effective information

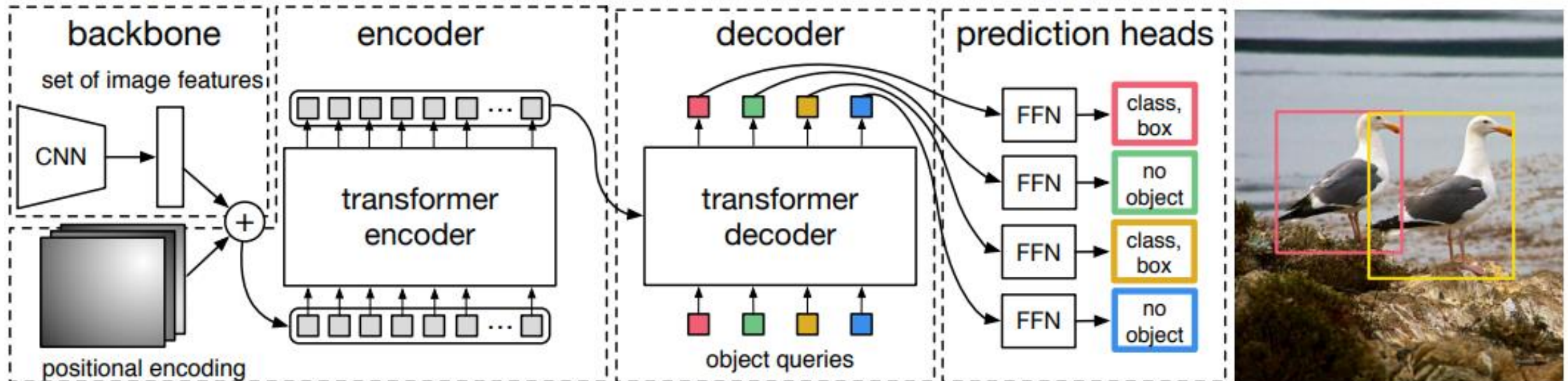
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- **DETR (DEtection TRansformer) series**



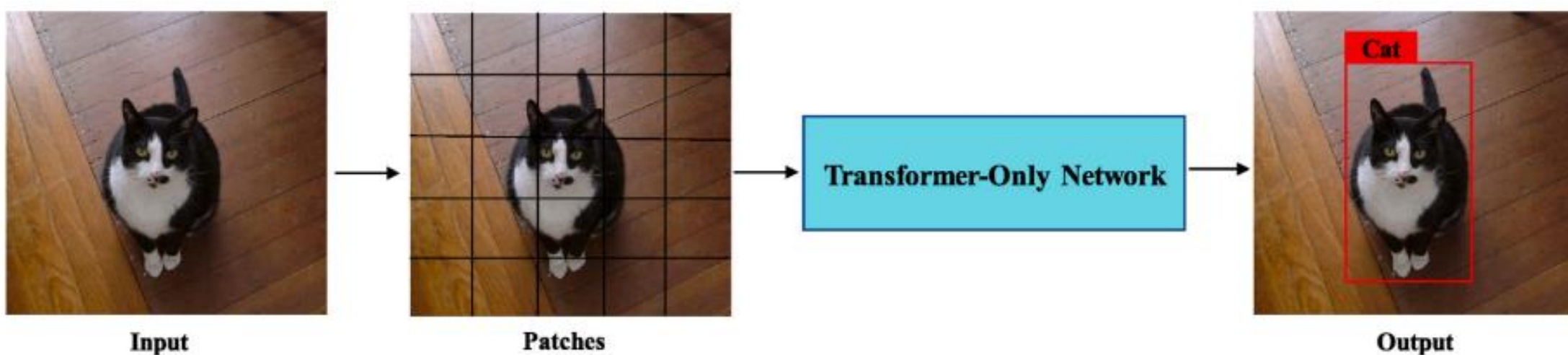
11.2 Machine learning based methods - Transformer-based methods

- CNN backbone to learn a 2D representation of an input image
- Flattens it and supplements it with a positional encoding before passing it into a transformer encoder
- A transformer decoder then takes as input a small fixed number of learned positional embeddings, which we call object queries, and additionally attends to the encoder output



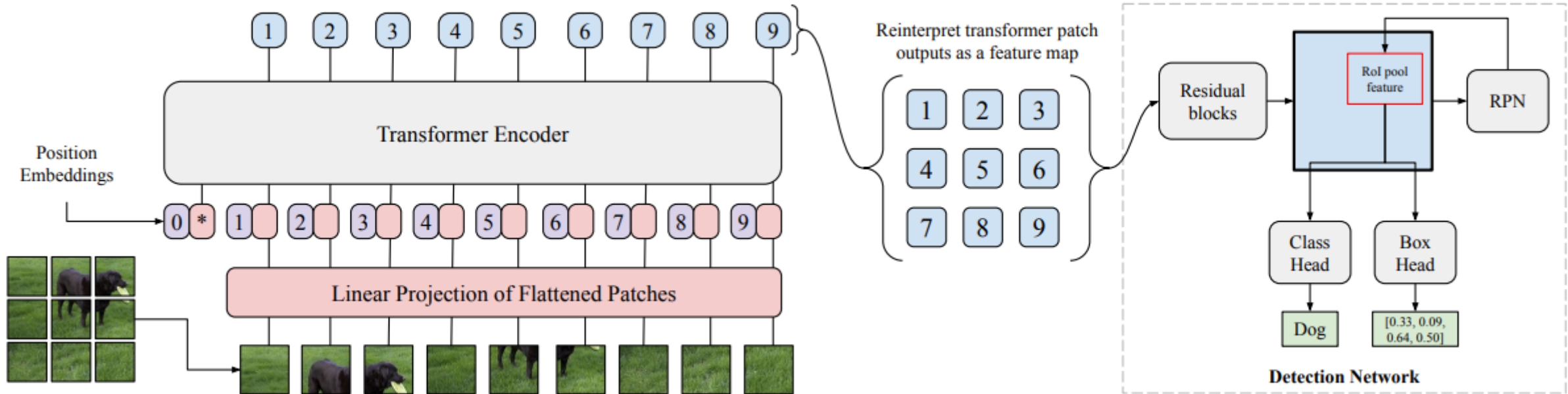
11.2 Machine learning based methods - Transformer-based methods

- ViT (Vision Transform) series



11.2 Machine learning based methods - Transformer-based methods

- ViT-FRCNN



Summary

- Fundamentals of object detection
- Traditional methods:
- Machine learning based methods